

ELIMUX MASTER RUNBOOK

Single Source of Truth | Last Updated: 2026-08-16

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1. PROJECT OVERVIEW

ElimuX is a global education discovery SaaS platform with two core pillars:

- **Course Discovery** — AI-powered search for scholarships, programs, institutions
- **Internship & Attachment** — University students (pre-graduation) and graduates (post-graduation)

Revenue features (partner ads, sponsorships) are secondary to core pillars.

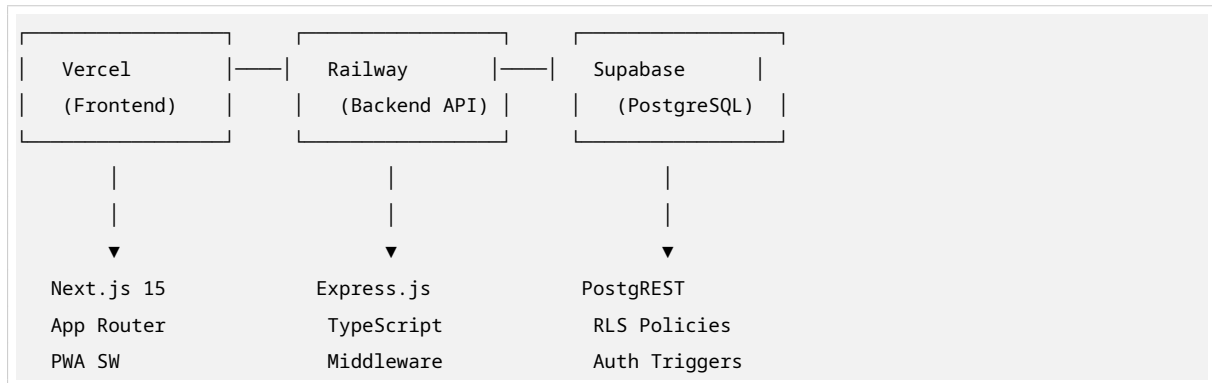
Stack:

- Frontend: Next.js 15 (App Router), TypeScript, Tailwind CSS, deployed on Vercel
- Backend: Express.js, TypeScript, deployed on Railway
- Database: Supabase (PostgreSQL + PostgREST)
- Payments: M-Pesa (Daraja API) + Stripe + Paystack
- AI: Multiple providers (Anthropic, OpenAI, DeepSeek, Together, Kimi)
- PWA: Service Worker with offline caching

Repos:

- `elimux-frontend` — Next.js app
 - `elimux-backend` — Express API
 - `elimux-sql` — Database migrations
 - `elimux-phases-23` — Project documentation
-

2. ARCHITECTURE



Critical Rule: Frontend is display-only. All writes go through backend API.

3. DATABASE SCHEMA

3.1 Core Tables

scholarships

Column	Type	Nullable	Notes
id	uuid	NO	PK
title	varchar	NO	
provider	varchar	NO	Free-text provider name
provider_logo_url	varchar	YES	
description	text	YES	
eligibility	text	YES	
benefits	text	YES	
amount	varchar	YES	
currency	varchar	YES	Default 'KES'
coverage_type	varchar	YES	
institution_id	uuid	YES	FK → institutions.id
country_id	uuid	YES	FK → countries.id
study_levels	array	YES	
disciplines	array	YES	
target_groups	array	YES	
application_opens	timestampz	YES	
application_deadline	timestampz	NO	
notification_date	timestampz	YES	
application_url	varchar	YES	
application_process	text	YES	
required_documents	array	YES	
status	varchar	YES	Default 'active'
is_featured	boolean	YES	Default false
source_url	varchar	YES	
scraped_at	timestampz	YES	

Table 1 – continued

Column	Type	Nullable	Notes
verified_at	timestampz	YES	
created_at / updated_at	timestampz	YES	Default now()
provider_id	uuid	YES	FK → scholar-ship_providers.id
sponsor_id	uuid	YES	FK → scholar-ship_sponsors.id
funding_amount	numeric	YES	
duration / duration_unit	int / text	YES	
is_sponsored	boolean	YES	Default false
created_by	uuid	YES	
application_status	text	YES	‘open’ checked by cron
summary,	text	YES	
cover_image_url			
amount_min,	numeric	YES	
amount_max			
announcement_date,	timestampz	YES	
program_start_date			
duration_months	int	YES	
location_type, country,	text	YES	country is free-text,
city			separate from
			country_id
eligibility_criteria	jsonb	YES	Default {}
application_method,	text	YES	
application_instructions			
view_count	int	YES	Default 0
tags, education_level,	array	YES	Default {}
field_of_study			

scholarship_providers

Column	Type	Notes
id	uuid	PK
name	varchar	
slug	varchar	
logo_url	varchar	
website	varchar	
description	text	
email	varchar	
phone	varchar	
verified	boolean	
status	varchar	Default ‘active’
is_partner	boolean	Default false
claimed_by	uuid	

Table 2 – continued

Column	Type	Notes
claimed_at	timestampz	
created_at / updated_at	timestampz	

RLS: Public SELECT (true), admin/super_admin ALL. **Status:** 0 rows as of 2026-08-16.

scholarship_sponsors

Column	Type	Notes
id	uuid	PK
name	varchar	
type	varchar	
logo_url	varchar	
website	varchar	
country_code	varchar	
contact_email	varchar	
contact_phone	varchar	
description	text	
is_verified	boolean	
tier	varchar	Default ‘free’
created_by	uuid	
created_at / updated_at	timestampz	

RLS: Public SELECT verified, admin ALL.

scholarship_applications

Column	Type	Nullable	Notes
id	uuid	NO	PK
student_id	uuid	NO	No FK constraint
scholarship_id	uuid	NO	FK → scholarships.id
status	text	YES	Default ‘draft’
documents_uploaded	jsonb	YES	Default []
missing_documents	jsonb	YES	Default []
ai_match_score	int	YES	
ai_guidance	text	YES	
deadline_reminder_sent	boolean	YES	Default false
reminder_dates	jsonb	YES	Default []
notes	text	YES	
created_at / updated_at	timestampz	YES	
applicant_id	uuid	YES	UNUSED – student_id is the active column

Table 4 – continued

Column	Type	Nullable	Notes
personal_statement	text	YES	
answers	jsonb	YES	
documents	jsonb	YES	
current_institution	text	YES	
current_level	text	YES	
field_of_study	text	YES	
gpa	text	YES	
review_score	int	YES	
review_notes	text	YES	
reviewed_by	uuid	YES	
reviewed_at	timestampz	YES	
submitted_at	timestampz	YES	

RLS: Students own ALL, Users insert own, Users update own draft, Admin ALL, Admin review UPDATE, SELECT for applicant or scholarship admin.

scholarship_profiles

Column	Type	Notes
id	uuid	PK
user_id	uuid	NOT NULL
gpa	numeric	
course_field	text	
study_level	text	
year_of_study	int	
county	text	
country_code	varchar	Default 'KE'
gender	text	
age	int	
financial_need	boolean	
orphan_status	boolean	
disability	text	
work_experience_years	int	
career_goals	text	
extracurriculars	array	
languages	array	
last_matched_at	timestampz	
created_at / updated_at	timestampz	

student_profiles

Column	Type	Notes
id	uuid	PK
user_id	uuid	
full_name	text	
email	text	
phone	text	
registration_number	text	
university_name	text	
course_name	text	
course_category	text	
year_of_study	int	Default 1
preferred_locations	array	
preferred_industries	array	
skills	array	
portfolio_url	text	
linkedin_url	text	
github_url	text	
resume_url	text	
is_university_verified	boolean	Default false
is_open_to_remote	boolean	
is_open_to_relocation	boolean	
nita_number	text	
created_at / updated_at	timestampz	
student_type	text	
graduation_year	int	

RLS: Own manage, Admin manage, Public read verified.

user_roles

Column	Type	Notes
id	uuid	PK
user_id	uuid	NOT NULL
role	text	NOT NULL (admin, super_admin)
created_at	timestampz	

RLS: Users view own role only. Role assignment via service-role key backend.

admin_users

Column	Type	Notes
id	uuid	PK
email	varchar	
role	varchar	Default 'editor'

Table 8 – continued

Column	Type	Notes
is_active	boolean	Default true
last_login	timestampz	
created_at	timestampz	

No RLS policies. Access is service-role only via x-admin-key.

3.2 Supporting Tables

- `scholarship_eligibility` — criteria per scholarship
- `scholarship_documents` — required docs per scholarship
- `scholarship_favorites` — device-scoped (not user-scoped)
- `scholarship_alerts` — keyword alerts by device/email
- `scholarship_messages` — messaging between parties

4. API ROUTES INVENTORY

4.1 Public Routes

Route	File	Auth
GET /api/scholarships	scholarships.ts	None
GET /api/scholarships/ featured	scholarships.ts	None
GET /api/scholarships/ favorites	scholarships.ts	None
GET /api/scholarships/:id	scholarships.ts	None
POST /api/scholarships/:id/ favorite	scholarships.ts	None
DELETE /api/ scholarships/:id/favorite	scholarships.ts	None
POST /api/scholarships/alerts	scholarships.ts	None
POST /api/scholarships/match	scholarship-matching.ts	None
GET /api/auth/me	auth.ts	None

4.2 Authenticated Routes (requireUser)

Route	File	Notes
GET /api/scholarships/match/me	scholarship-matching.ts	
POST /api/ scholarship-applications/start	scholarship-applications.ts	Body: { scholarship_id }
GET /api/scholarship-applications	scholarship-applications.ts	Own applications
GET /api/ scholarship-applications/:id	scholarship-applications.ts	

Table 10 – continued

Route	File	Notes
POST /api/ scholarship-applications/:id/upload	scholarship-applications.ts	Base64 file upload
POST /api/ scholarship-applications/:id/submit	scholarship-applications.ts	
POST /api/ scholarship-applications/:id/ guidance	scholarship-applications.ts	Body: { force? }
DELETE /api/ scholarship-applications/:id	scholarship-applications.ts	
POST /api/ scholarship-providers/:id/claim	scholarship-providers.ts	Any authenticated user

4.3 Admin Routes (adminMiddleware)

Route	File	Notes
GET /api/admin/scholarships	admin-scholarships.ts	
GET /api/admin/scholarships/:id	admin-scholarships.ts	
PUT /api/admin/scholarships/:id	admin-scholarships.ts	
DELETE /api/admin/ scholarships/:id	admin-scholarships.ts	
POST /api/admin/scholarships	admin.ts	Live creation endpoint
PATCH /api/admin/ scholarships/:id	admin.ts	
GET /api/admin/ scholarship-sponsors	admin-scholarship-sponsors.ts	
POST /api/admin/ scholarship-sponsors	admin-scholarship-sponsors.ts	
GET /api/admin/ scholarship-applications	admin-scholarship- applications.ts	Partner-only filter
POST /api/admin/ scholarship-applications/:id/review	admin-scholarship- applications.ts	
POST /api/admin/ scholarship-applications/ send-reminders	admin-scholarship-reminders.ts	Manual trigger
POST /api/admin/ scholarship-providers/:id/ approve-partnership	scholarship-providers.ts	
GET /api/admin/ scholarship-providers	scholarship-providers.ts	Filters: claimed_only, partner_only
GET /api/admin/scrapper/ scholarships/changes	admin-scraper-scholarships.ts	
POST /api/admin/scrapper/ scholarships/run	admin-scraper-scholarships.ts	

Table 11 – continued

Route	File	Notes
POST /api/admin/scrapper/scholarships/changes/:id/approve	admin-scraper-scholarships.ts	
POST /api/admin/scrapper/scholarships/changes/:id/reject	admin-scraper-scholarships.ts	
GET /api/admin/scrapper/scholarships/sources	admin-scraper-scholarships.ts	
GET /api/admin/dashboard	admin-dashboard.ts	
GET /api/admin/analytics	admin-analytics.ts	
GET /api/admin/reports	admin-reports.ts	
GET /api/admin/settings	admin-settings.ts	
GET /api/admin/payments	admin-payments.ts	
POST /api/admin/users	auth.ts	Uses adminAuth (not adminMiddleware)
PATCH /api/admin/users/:id/role	auth.ts	Uses adminAuth
PATCH /api/admin/users/:id/status	auth.ts	Uses adminAuth
DELETE /api/admin/users/:id	auth.ts	Uses adminAuth

4.4 Cron Routes

Route	File	Auth
POST /api/cron/scholarship-reminders	cron-scholarship-reminders.ts	x-cron-secret header

Infra: Railway cron service scholarship-reminders-cron, runs daily at 05:00 UTC.

5. AUTHENTICATION & AUTHORIZATION

5.1 Auth Middleware

Middleware	Location	Type	Used By
requireUser	middleware/user-auth.ts	Async	scholarship-applications, scholarship-matching
adminMiddleware	middleware/auth.ts	Sync	Most admin routes
adminAuth	middleware/auth.ts	Async	routes/auth.ts (users CRUD)
authenticateToken	middleware/rbac.ts	Async	RBAC system
requireRole	middleware/rbac.ts	Async	Role-specific routes
institutionAuth	middleware/institution-auth.ts	Async	Institution portal
employerAuth	middleware/employer-auth.ts	Async	Employer portal

Table 13 – continued

Middleware	Location	Type	Used By
advertiserAuth	middleware/ advertiser-auth.ts	Async	Advertiser portal

⚠ SECURITY NOTE: `adminAuth` and `adminMiddleware` are NOT confirmed identical. `adminAuth` is async and may do more than header comparison. `adminMiddleware` is sync and only checks `x-admin-key` header. Replacing one with the other would change behavior. **This is being documented in Cycle 003.**

5.2 Frontend Auth

Context: `src/context/AuthContext.tsx`

- Exports: `AuthProvider`, `useAuthContext()`
- Returns: `{ user, session, loading, isAuthenticated, signIn, signUp, signOut, refreshUser }`
- Singleton client: `src/lib/supabase/client.ts` (memoized, prevents Web Locks deadlock)

⚠ CRITICAL: Never call `supabase.auth.getSession()` directly in a component. Always consume `useAuthContext()`. Direct calls can deadlock via Web Locks API.

Hook alias: `src/hooks/useAuth.ts` re-exports `useAuthContext` as `useAuth`. Used only in `DesktopNav.tsx`. Everywhere else imports `useAuthContext` directly.

5.3 Admin Key Flow

1. Admin logs in → frontend stores admin key in `AdminKeyContext`
2. Admin pages read key from context
3. `lib/api.ts` passes key as 3rd arg to `request()`, sets `x-admin-key` header
4. Backend `adminMiddleware` validates header against `process.env.ADMIN_KEY`

6. ENVIRONMENT VARIABLES

6.1 Backend (elimux-backend)

Variable	Purpose	Sensitivity
<code>ADMIN_KEY</code>	Admin API access	HIGH
<code>CRON_SECRET</code>	Cron endpoint protection	HIGH
<code>SUPABASE_SERVICE_ROLE_KEY</code>	Database admin access	CRITICAL
<code>SUPABASE_URL</code>	Database connection	MEDIUM
<code>STRIPE_SECRET_KEY</code>	Payments	CRITICAL
<code>PAYSTACK_SECRET_KEY</code>	Payments	CRITICAL
<code>MPESA_CONSUMER_KEY</code>	M-Pesa API	CRITICAL
<code>MPESA_CONSUMER_SECRET</code>	M-Pesa API	CRITICAL
<code>ANTHROPIC_API_KEY</code>	AI provider	HIGH
<code>OPENAI_API_KEY</code>	AI provider	HIGH

Table 14 – continued

Variable	Purpose	Sensitivity
DEEPSEEK_API_KEY	AI provider	HIGH
TOGETHER_API_KEY	AI provider	HIGH
KIMI_API_KEY	AI provider	HIGH
RESEND_API_KEY	Email service	HIGH
VAPID_PRIVATE_KEY	Push notifications	HIGH
VAPID_PUBLIC_KEY	Push notifications	MEDIUM
FRONTEND_URL	CORS/redirects	LOW
PORT	Server port	LOW
API_KEY / API_URL	Internal API	MEDIUM
AI_MODE /	AI configuration	LOW
AI_FALLBACK_ORDER		
SCRAPER_ALLOWED_DOMAINS	Scraper whitelist	LOW
RECEIPTS_FROM_EMAIL	Email sender	LOW
NEXT_PUBLIC_API_URL	Frontend API URL	LOW

6.2 Frontend (elimux-frontend)

Variable	Purpose	Sensitivity
NEXT_PUBLIC_SUPABASE_URL	Supabase connection	MEDIUM
NEXT_PUBLIC_SUPABASE_ANON_KEY	Supabase key	LOW
NEXT_PUBLIC_API_URL	Backend API URL	LOW
NEXT_PUBLIC_SITE_URL	Site URL	LOW
NEXT_PUBLIC_VAPID_PUBLIC_KEY	Push notifications	LOW
NEXT_PUBLIC_FEATURE_SKILLSTAG_TOGGLE	SkillsTag	LOW
NEXT_PUBLIC_FEATURE_SKOLEX_ADS	Skolex ADS	LOW
NEXT_PUBLIC_FEATURE_SKOLEX_HOME	Skolex HOME	LOW
ADMIN_KEY	Admin API access	HIGH
SUPABASE_SERVICE_ROLE_KEY	Service role (if used server-side)	CRITICAL

⚠ CRITICAL: NEXT_PUBLIC_ prefix makes variables visible in browser DevTools. Never prefix secrets with NEXT_PUBLIC_.

7. PWA / OFFLINE STATUS

7.1 Service Worker (public/sw.js)

- **Static cache:** elimux-static-v4 — cache-first for same-origin assets
- **API cache:** elimux-api-v4 — network-first for most GETs
- **Scholarship cache:** Stale-while-revalidate for /api/scholarships*
- **Excluded:** /api/scholarship-applications* stays network-first

7.2 Pre-cached Shell Pages

- / (home)
- /scholarships/
- /applications/

7.3 Background Sync

- **Hook:** `useBackgroundSync()` — mount exactly once via `BackgroundSyncManager.tsx`
- **Queueable actions:** favorite, review, application
- **Storage:** `localStorage` (`elimux-local-pending-actions`), NOT `IndexedDB`
- **⚠️ Limitation:** No document/file upload queue. Base64 files capped at 5MB. `localStorage` quota risk if files are queued.

7.4 Offline Indicator

- Component: `OfflineIndicator.tsx`
- Status: Now wired into root `layout.tsx` (added 2026-08-16)

8. KNOWN ANOMALIES & SECURITY NOTES

8.1 Resolved in Cycle 002

-  `scholarships.scholarship_provider_id` — DROPPED. Was dangling column with zero code references.

8.2 Resolved in Cycle 003

-  Dead POST / in `admin-scholarships.ts` — REMOVED. Was shadowed by `admin.ts`.

8.3 In Progress (Cycle 004)

-  `adminAuth` vs `adminMiddleware` discrepancy — Being documented with JSDoc.

8.4 Active Risks

Risk	Severity	Status
<code>scholarship_providers</code> has 0 rows	MEDIUM	All scholarships show discovery-only mode
No FK on <code>student_profiles.user_id</code>	MEDIUM	Data integrity not enforced at DB level
<code>adminAuth</code> vs <code>adminMiddleware</code> may differ	HIGH	Being documented
Document uploads have no offline queue	MEDIUM	PWA incomplete for file uploads
<code>scholarship_favorites</code> is device-scoped	LOW	Not user-scoped (no auth required)

Table 16 – continued

Risk	Severity	Status
Two competing scholarship creation endpoints	RESOLVED	admin.ts is live, admin-scholarships.ts POST removed

8.5 Naming Collision Risks

- `useAuth` (from hooks) vs `useAuthContext` (from context) — Import path matters
- `applicationApprovedEmailHtml` (institution apps) vs `scholarshipAwardedEmailHtml` (scholarship apps) — Different domains, similar names

9. CYCLE HISTORY

Cycle 000 — BASELINE

- **Date:** 2026-08-16
- **Trigger:** System initialization
- **Goal:** Create bridge infrastructure and ground truth audit
- **Status:** COMPLETE
- **Commits:** `bridge.md` created, §1–§9 audit preserved

Cycle 001 — BLOCKED

- **Date:** 2026-08-16
- **Trigger:** Instruction 001 (drop dangling column)
- **Goal:** Drop `scholarships.scholarship_provider_id`
- **Status:** BLOCKED — Instruction truncated in paste
- **Archive:** `bridge-001.md`

Cycle 002 — COMPLETE

- **Date:** 2026-08-16
- **Trigger:** Instruction 001 retry
- **Goal:** Drop `scholarships.scholarship_provider_id`
- **Status:** COMPLETE
- **Deviations:**
 1. `elimux-sql/migrations/` doesn't exist — used flat `elimux-sql/` instead
 2. `bridge-001.md` already existed — archived to `bridge-002.md` instead
- **Commits:**
 - `elimux-frontend`: `c862006..ba808bf`
 - `elimux-sql`: `ec92f10..9e870f9`
- **Migration:** `43_drop_dangling_scholarship_provider_id.sql`

Cycle 003 — COMPLETE

- **Date:** 2026-08-16
- **Trigger:** Instruction 002 (remove dead POST handler)

- **Goal:** Remove dead POST / from admin-scholarships.ts
- **Status:** COMPLETE
- **Deviations:** None
- **Commits:**
 - elimux-backend: ec92f10..9e870f9
 - elimux-frontend: c862006..ba808bf

Cycle 004 — IN PROGRESS

- **Date:** 2026-08-16
 - **Trigger:** Instruction 003 (document admin auth discrepancy)
 - **Goal:** Add JSDoc to adminAuth and adminMiddleware
 - **Status:** IN PROGRESS
-

10. ACTIVE WORK QUEUE

10.1 Immediate (This Session)

- Cycle 004: Document adminAuth vs adminMiddleware (IN PROGRESS)
- Cycle 005: Verify adminAuth and adminMiddleware are not interchangeable
- Cycle 006: Add offline queue for document uploads (if prioritized)

10.2 Short-Term (Next 1–2 Weeks)

- Populate scholarship_providers table or remove partner-gate until data exists
- Add FK constraint on student_profiles.user_id
- Implement proper RBAC enforcement in all admin routes
- Add rate limiting to public endpoints
- Security headers (CSP, X-Frame-Options, etc.)
- GDPR data export endpoint
- GDPR account deletion endpoint

10.3 Medium-Term (Next Month)

- M-Pesa integration completion
 - Stripe Connect for partners
 - PWA file upload queue with IndexedDB
 - Automated testing suite
 - Performance monitoring (Supabase slow query logs)
-

11. DEPLOYMENT NOTES

11.1 Frontend (Vercel)

- Build command: npm run build (Next.js)
- Environment variables set in Vercel dashboard
- Images: images.unoptimized: true (deferred to post-launch)

11.2 Backend (Railway)

- Build command: `npm run build` (TypeScript `tsc`)
- Service: `elimux-backend`
- Cron service: `scholarship-reminders-cron` (separate Railway service)

11.3 Database (Supabase)

- Project: `ohlgjvenwekpbpykutz`
- Migrations: Flat files in `elimux-sql/` (not `migrations/` subfolder)
- RLS: Active on all tables
- PITR: Enable for production (\$25/month add-on)

11.4 Git Workflow

- Branch protection on `main`
- All changes via Pull Request
- Commit convention: `cycle-NNN: description`
- Co-authored-by: Claude for backend changes

12. EMERGENCY PROCEDURES

12.1 Rollback a Bad Cycle

```
# Find the cycle number in audit-log.md
cp docs/archive/bridge-YYY.md docs/bridge.md
git add docs/bridge.md
git commit -m "rollback: restored bridge.md to cycle-YYY"
```

12.2 Rotate Leaked Secret

1. Generate new key in provider dashboard (Stripe, Supabase, etc.)
2. Update in Railway/Vercel environment variables
3. Delete old key from provider dashboard
4. Check git history: `git log -p | grep OLD_KEY`
5. If committed, rotate immediately and consider history rewrite

12.3 Database Emergency

- **PITR restore:** Supabase Dashboard → Database → Backups → Point in Time Recovery
- **Manual rollback:** Run inverse migration from `elimux-sql/`
- **Contact:** Supabase support for unrecoverable issues

12.4 API Down

- Check Railway dashboard for service status
- Check Supabase dashboard for DB connectivity
- Verify environment variables are set (especially `SUPABASE_SERVICE_ROLE_KEY`)
- Check recent commits for breaking changes

APPENDIX A: FILE LOCATIONS

Purpose	Path
Bridge file	elimux-frontend/docs/bridge.md
Audit log	elimux-frontend/docs/audit-log.md
Archive	elimux-frontend/docs/archive/bridge-NNN.md
Claude prompt	elimux-frontend/claude-prompt.txt
This runbook	elimux-frontend/docs/ ELIMUX_MASTER_RUNBOOK.md
Backend routes	elimux-backend/src/routes/
Backend middleware	elimux-backend/src/middleware/
Frontend auth	elimux-frontend/src/context/AuthContext.tsx
Frontend API client	elimux-frontend/src/lib/api.ts
Supabase client	elimux-frontend/src/lib/supabase/client.ts
Service worker	elimux-frontend/public/sw.js
Migrations	elimux-sql/ (flat)

*This document is append-only for history sections. Core sections may be updated as the project evolves.
Last updated by: Kimi, 2026-08-16.*